

# Problem Statement



## Scale Problem

Online news volume far exceeds human fact-checking capacity

## Mimicry Problem

Fake articles deliberately mimic legitimate journalism in tone and style

## Speed Problem

Misinformation spreads faster than corrections can be published

## Access Problem

No widely accessible tool exists for everyday users to verify credibility

# Business Objectives



**01**

## High-Accuracy Classifier

Build a reliable binary ML system distinguishing fake from real news

**02**

## User-Friendly Web Interface

Any user can paste an article and get an instant prediction with confidence score

**03**

## Demonstrate Classical ML

Show TF-IDF + Logistic Regression effectiveness on NLP tasks

**04**

## Reproducible Pipeline

Well-documented, open-source ML pipeline for academic & future use

# Dataset Profile



## Source

Kaggle — Clément Bisailon  
Fake and Real News Dataset  
44,898 total articles

**23,481**

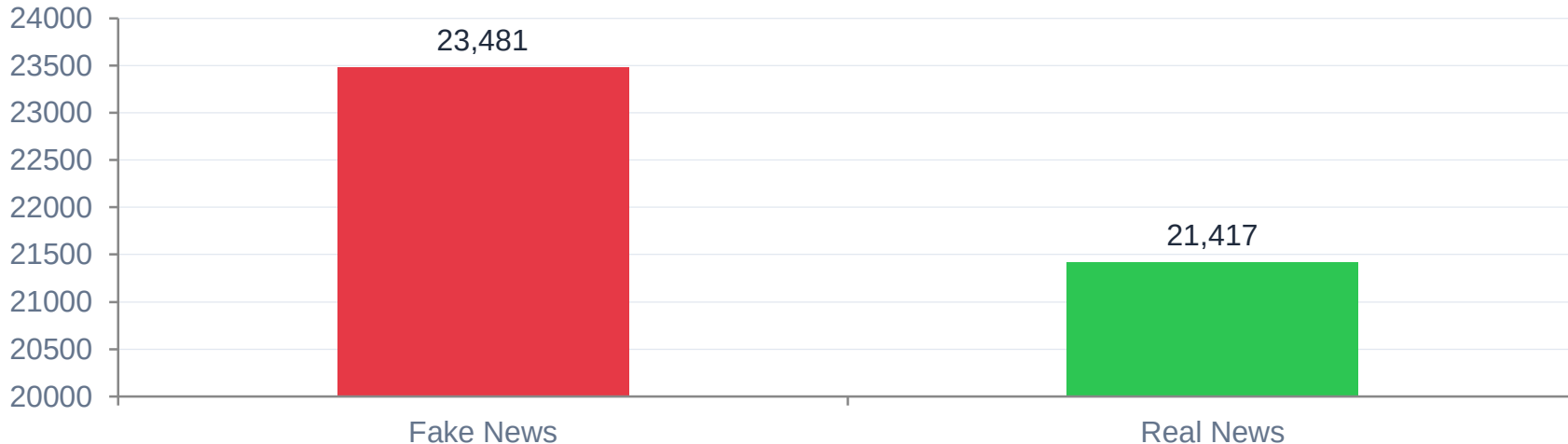
Fake Articles

**21,417**

Real Articles

**2015–18**

Time Period



# Pipeline Architecture



## 01 Data Ingestion

Load Fake.csv + True.csv → assign labels 0/1 → shuffle & merge

## 02 Text Preprocessing

Concatenate title + body → regex clean → lowercase → stopword removal → Porter Stemming

## 03 TF-IDF Vectorization

Convert cleaned text to numerical features (top 5,000 terms)

## 04 Train / Test Split

80% training / 20% test with random\_state=42 for reproducibility

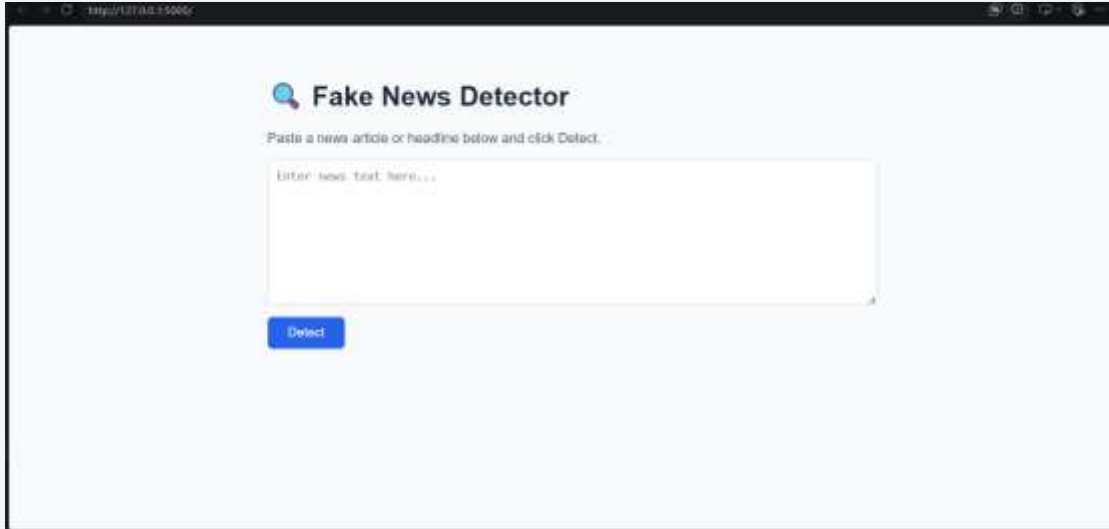
## 05 Model Training

Logistic Regression classifier fit on TF-IDF feature matrix

## 06 Deployment

Serialize model.pkl + vectorizer.pkl → serve via Flask web app

# Web Application Interface



The screenshot shows a web browser window with the address bar displaying 'http://127.0.0.1:5000/'. The page content includes a title 'Fake News Detector' with a magnifying glass icon. Below the title is a subtitle 'Paste a news article or headline below and click Detect.' and a large text input area with the placeholder text 'Enter news text here...'. A blue 'Detect' button is positioned below the input area.

## Text Input Area

Paste any news headline or full article text

## Detect Button

Triggers POST request to Flask /predict endpoint

## Live Result

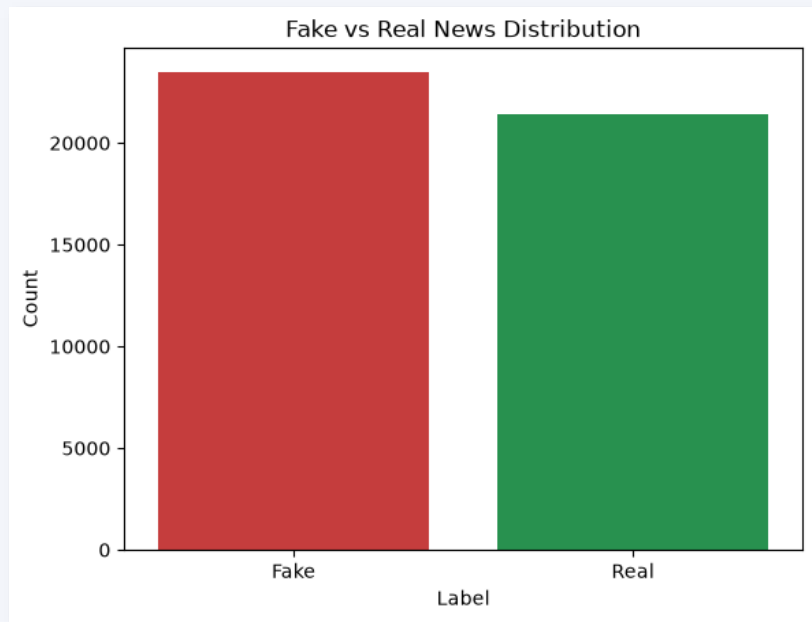
Shows FAKE ✗ or REAL ✔ with % confidence inline

## No Page Refresh

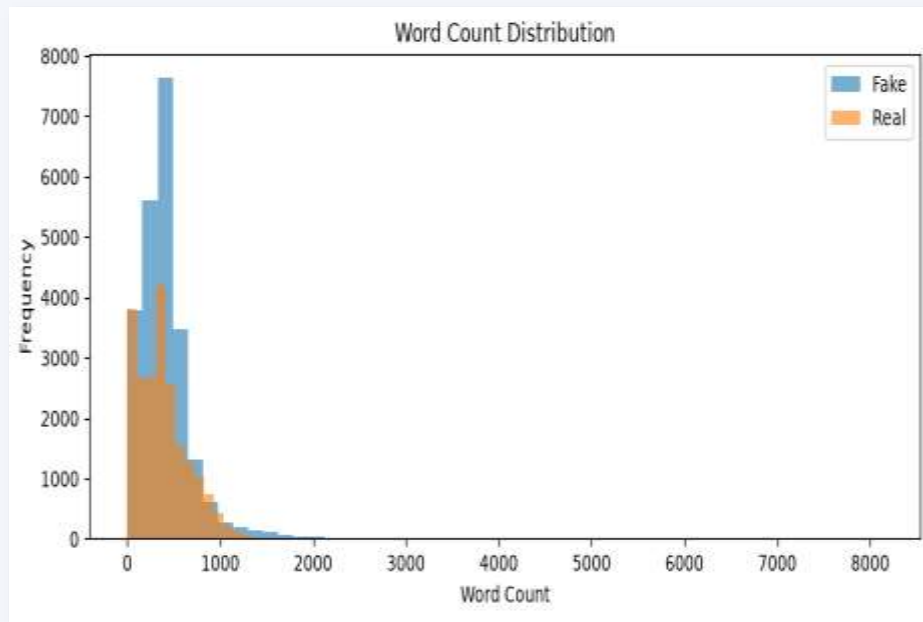
JavaScript fetch API updates result without reload

*localhost:5000 → python app.py*

# Exploratory Data Analysis

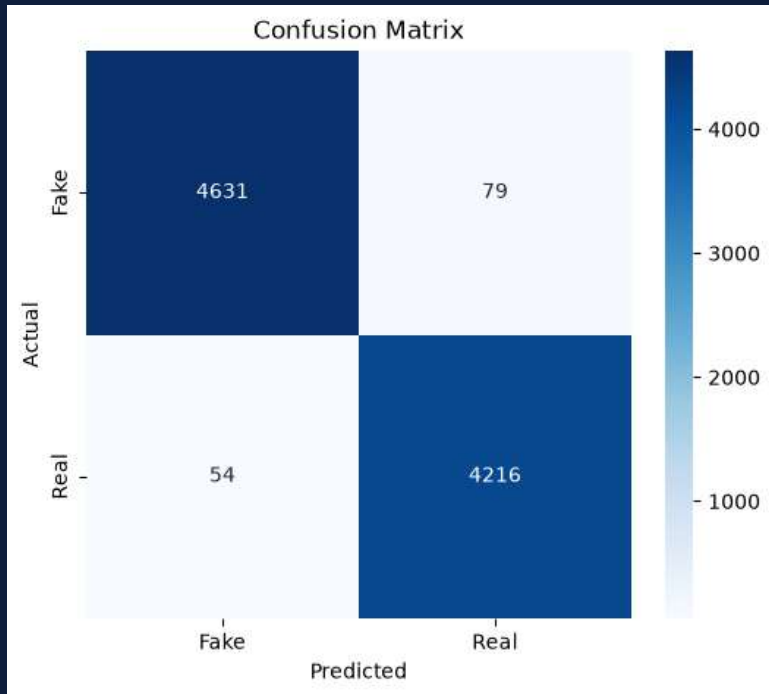


*Fig 1 — Label Distribution*  
*Near-balanced dataset prevents class bias*



*Fig 2 — Word Count Distribution*  
*Fake news articles tend to be longer*

# Model Evaluation



*Confusion Matrix — 20% held-out test set*

**98.3%**

Overall Accuracy

**4,631**

True Negatives (Fake → Fake)

**4,216**

True Positives (Real → Real)

**79**

False Positives

**54**

False Negatives

# Limitations & Future Work



## Current Limitations

- Domain Sensitivity — trained on 2015-18 political news only
- Context Blindness — TF-IDF cannot capture sarcasm or tone
- Static Vocabulary — new slang/terms unrecognised at inference

## Future Work

- Migrate to BERT / RoBERTa for semantic understanding
- Expand dataset across health, sports & entertainment news
- Add URL-based input — paste a link, not the article text
- DBSCAN anomaly detection for borderline articles



# Conclusion



98.3%

Overall Accuracy

- ✓ TF-IDF + Logistic Regression achieves high accuracy on news classification
- ✓ Complete end-to-end pipeline: ingestion → training → evaluation → web deployment
- ✓ Flask app makes the model accessible to any user via browser
- ✓ Open-source & reproducible — publicly available on GitHub

Thank You